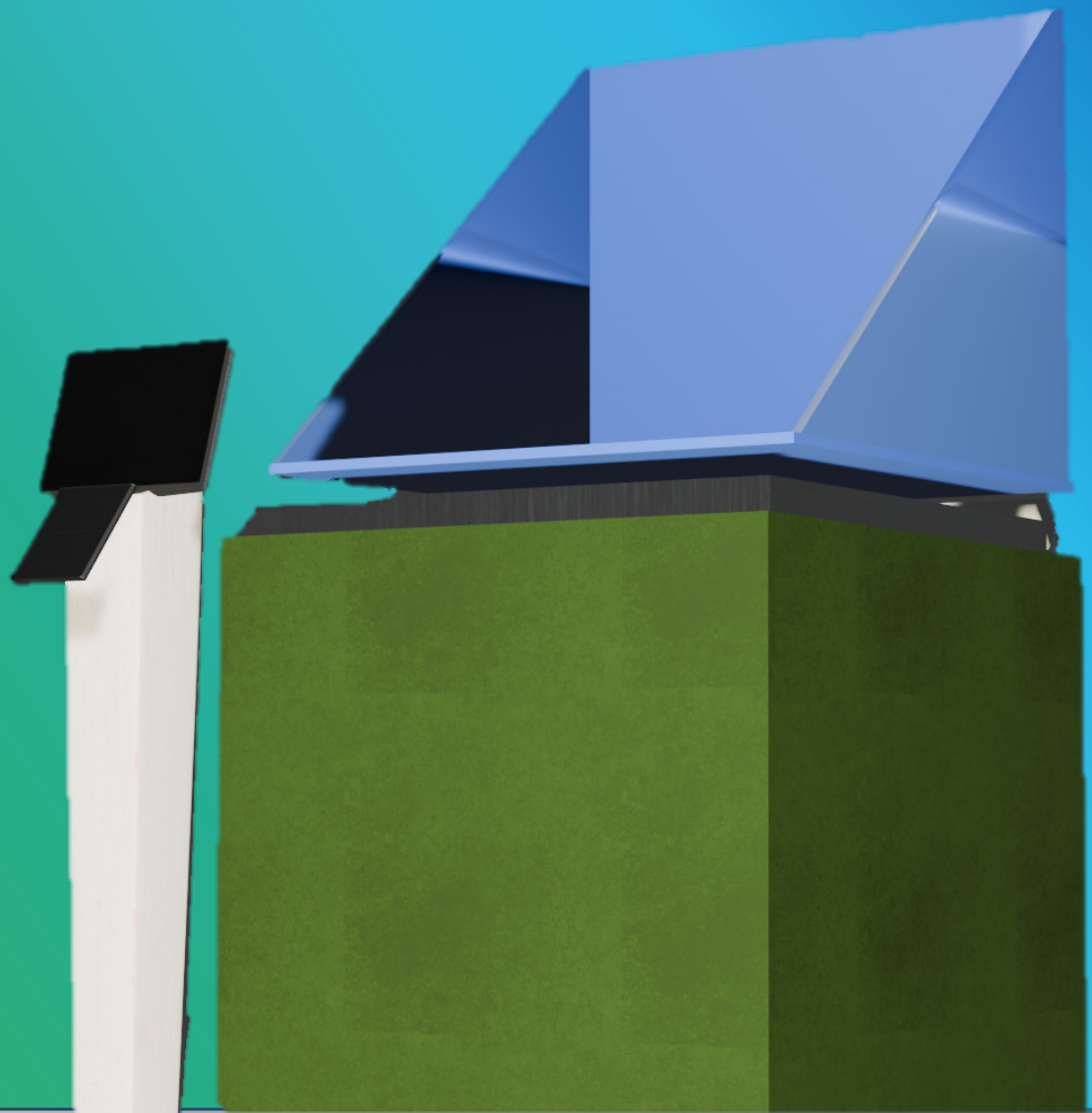
ACD Final Report

TU Delft

avalex



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Introduction

Vision

Avalex has a strong vision where they want to achieve a clean and circular living environment through responsible waste collection and raw materials management aiming to reduce residual waste by half every 5 years.

Mission

'Improving the knowledge and awareness of customers about the reusability and separation of products and materials, with the use of a rewarding system.'

Market gap

Avalex competes with other waste management companies on a market whith a limited capacity. Therefore, when waste is separated more correctly at Avalex, it will improve their position in this market.
The waste processing prices have been increased, so when the collected waste requires less post-separation, it will be more atractive for HVC to process waste from Avalex (3).

Project statement

Our project statement given to us as group A is: From customer to recycling centre, improve the customer experience on the different Avalex recycling centers. Achieve a new circular level by improving the appearance, user-friendliness, and educational value within the boundaries of the existing locations.

Multiple sudies have been executed:
- A cultural research to identify all cultural frustrations and possibilities to fix them.
- An ergonomical study to evaluate which design is physically preferred.
- A technological design to test which mechanism works best.

Uniqueness

The design is focused on changing peoples behaviour, through different sources.
The final design contributes to all selected drivers and Avalex's vision.

Concept drivers

After the cultural research (2) several frustrations have been identified amongst Avalex customers. Four of these frustrations were used as concept drivers:



Summary

Sustainable movement

To achieve Avalex's goals and meet their vision, something has to be changed about the way Avalex's customers act on the locations.
Therefore, we must first change the attitude of the customers towards waste.
With this project, I want Avalex to be something more than a place where people come and go as quickly as possible, tackling current frustrations and motivating customers to separate more correctly.

Target group

With my vision and mission I try to inform as many people as possible. I am focused on all Avalex visitors (aged 20-94). Which only excludes people that do not visit Avalex at all. People can receive feedback about their actions once they have visited one of the Avalex locations.

Stakeholders

Customers gain more knowledge about the consequences of their own actions and get rewarded after their visit.

Avalex gains more organised data and better separated waste. They also improve customers' attitude towards behaviour. They can also save money through less contamination and a lower need for employees helping customers.

The municipality gets data about separated waste per household.

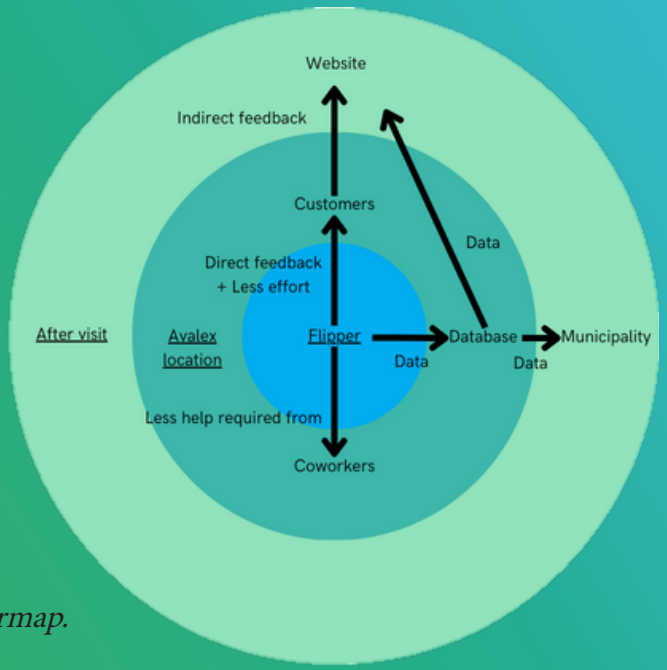


Figure 1: Stakeholdermap.

Societal impact

From the cultural research, I found that to change the behaviour of Avalex's customers, we must boost their self-efficacy by giving direct feedback about their actions at the locations(5).

By indirect feedback after visiting the locations, through the website with saved emisisons, rewards and savings on the Diftar system, the goal is to improve customers' perception behaviour control (4).

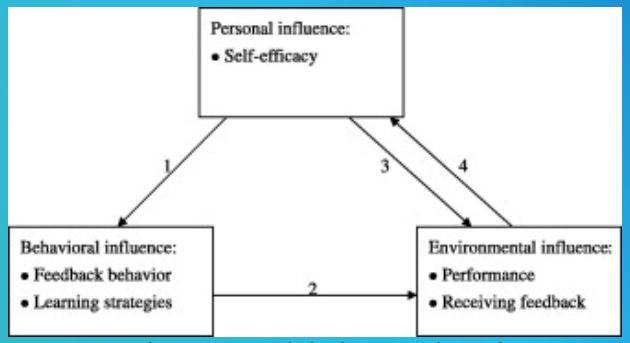


Figure 2: The personal, behavioral, and environmental influences of the social cognitive model.

This will eventually result in a more frequent visit to Avalex, which will change their habits towards more pro-environmental behaviour (2).

Criteria

For this project to succeed, a list of criteria has been made where the final design should comply with.

1. The design must be of limited scale, applicable at the Avalex locations.

2. The final design should be less than the given budget of 50.000 euro.

3. The design must require minimal effort from users to throw away their waste.

4. The design should be directly applicable.

5. The design must give back relevant feedback to the users.

6. The design must motivate customers to come to Avalex.

7. The design must be accessible for all customers to use.

8. The scaling platform should have the option to calibrate.

9. The interface of the design should be understandable and easy to use.

10. The design should be safe for customers to use.

11. The design should be easy to clean and maintain.

12. Before registration, users must have been given a consent form with information to decide if they want their data registered in return for feedback and rewards.

The Flipper

The main idea of the Flipper is that people will learn more about waste separation because of direct feedback from the Flipper at the Avalex location, and indirectly from helptools at the website showing impact, tutorials and more.

The Fliper is installed at the side of the standing platform at the Avalex locations, just before the fence. It's metal casing is connected to the hatch opening of a closed container.

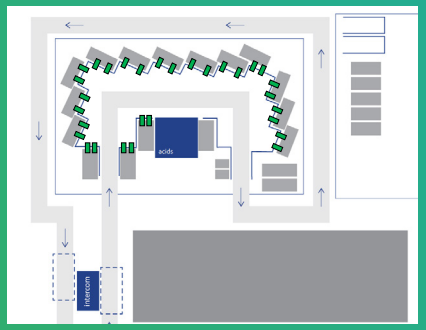


Figure 3: The Flippers at the Rijswijk location.

The height of the weighing platform where the waste is disposed on is 63 cm.
This height has been chosen through an ergonomical research which concluded this as the preferred height for heavy weights (appendix A).

The mechanism of the Flipper is installed with a rotation motor. The blue Flipper is fastened with a hinge to the supporting platform beneath, chosen through a technical study (appendix B).

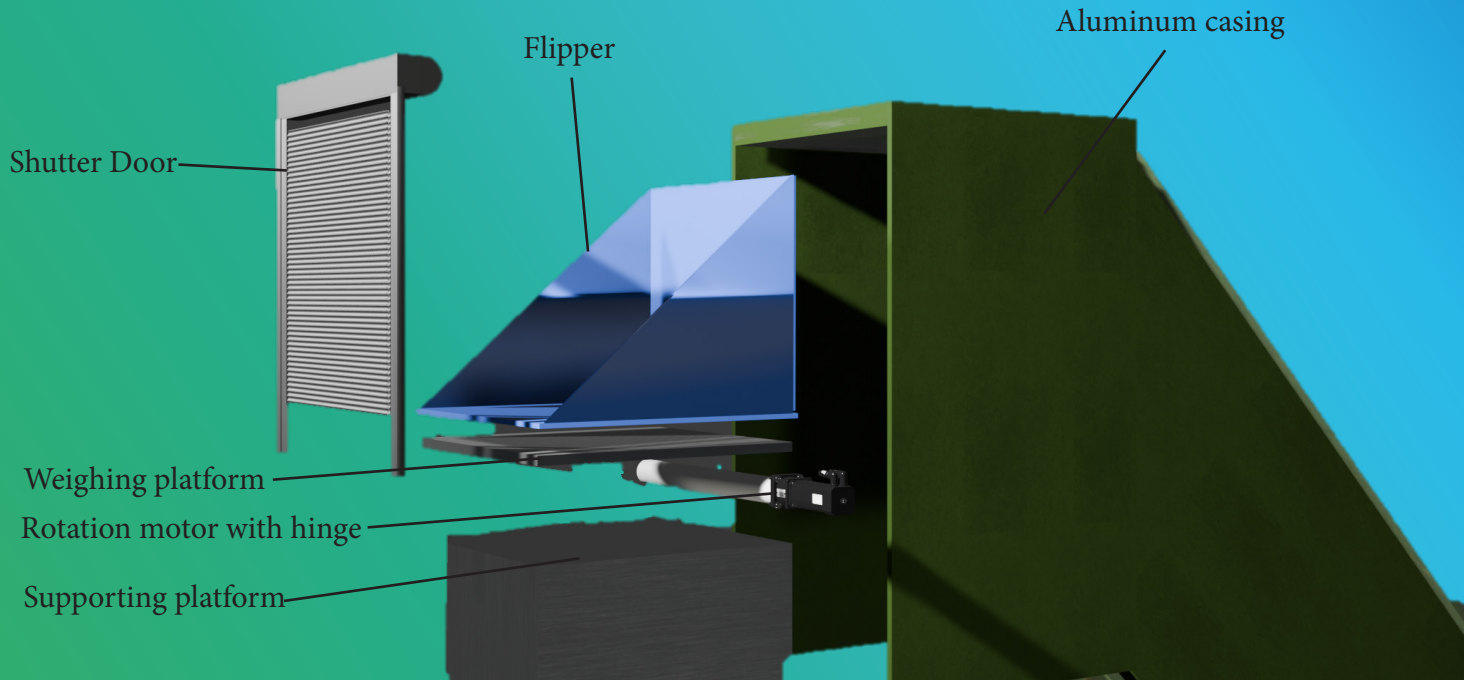
Context

The design is made to be directly appliccable on the Avalex locations. The design consists of a shutter door, weighing platform, the Flipper, a rotation motor, an interface with NFC reader and a green metal casing including a slide down the container.

Before the Flipper is operational, electricity cables or outlets must be laid to provide power for the motor, door and interface to be functional.



Figure 4: Flippers in context.



Usability

To make sure that customers cannot dispose their waste without using the Flipper, a new container is used with a closed roof, only letting in waste through two openings at the top which are connected to the casing of the flipper.



Figure 5: New container

When arriving at Avalex, customers can bring their waste towards the correct container.
When all waste is put in the Flipper, the customer can choose between a registered disposal by applying the Avalex pass on the NFC reader at the intercom or an unregistered one by pressing the button.

After scanning the Avalex pass or pressing the button, the shutter will close and the weight will be shown at the intercom and either be registered on the customers' account or not.

Interface

The interface is made out of a weather proof touchscreen monitor showing the amount of waste that is thrown away, emissons saved by the customer, a consent message where the use of the Avalex pass is explained and for customers who do not accept the registration of their waste, a disposal button. The NFC reader is placed below the interface.

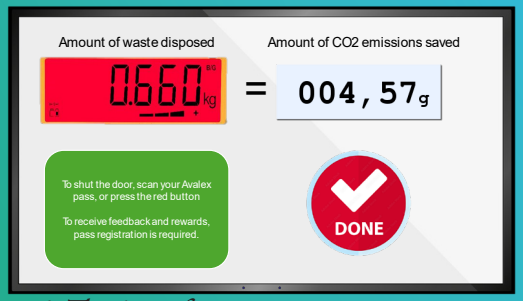


Figure 6: The interface.

The NFC chip will be implemented in the Avalex pass, which communicates the avalex ID number of the household to the NFC reader in the interface.

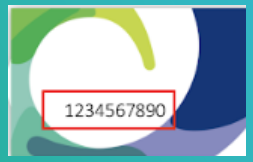
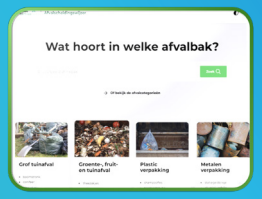


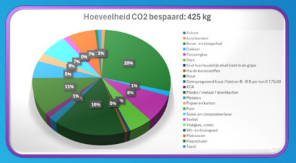
Figure 7: The Avalex pass.

Website

After visiting Avalex, the customers can go to the website, where they can check the following feedback:



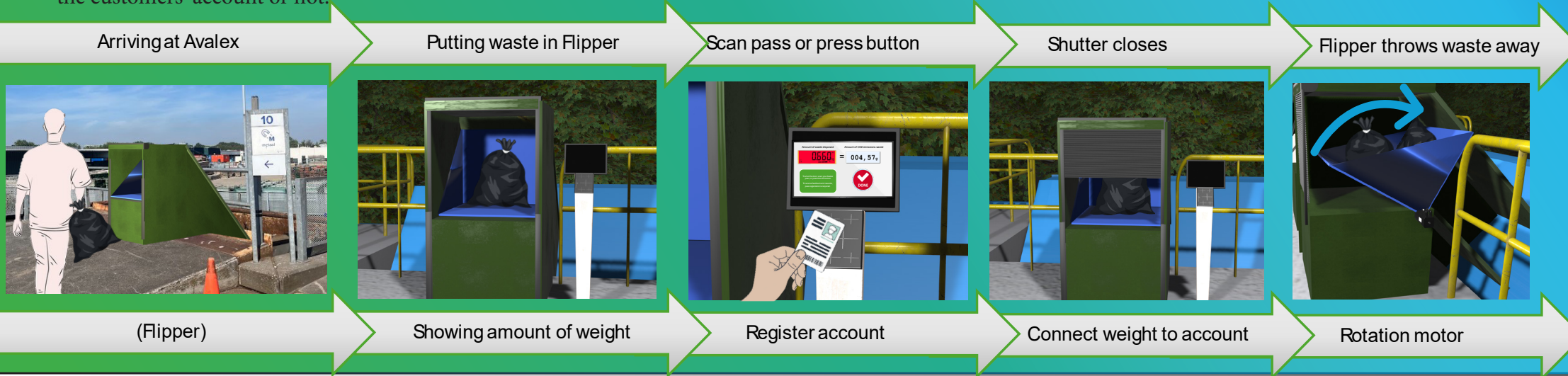
Where to throw away which materials at the Avalex location, how they can separate and recycle at home, and what happens to the waste after collected at Avalex.



How much emissions they have saved by separating their waste at Avalex instead of disposing it at the residual waste.
For these caluculations, each Flipper is connected to a type of material.



How much money they have saved with the Diftar-system by separating their waste instead of producing residual waste.
This is te reward the customers receive by visiting Avalex.



Desirability

To answer the question if the Flipper is desirable, let's take a look at the advantages the design brings to the stakeholders.

For Avalex, it is advantageous that the waste will be better sorted due to the gain in knowledge of the customers by the personal feedback and tutorial videos that they can receive online after their visit. Avalex can also receive data from the amounts of different kinds of waste they receive from their customers. Which can help them optimise their management of the different locations. Also keep in mind that their market position will be improved.

For Avalex' coworkers it is pleasant that they do not need to help customers to lift their heavy waste above the fence before disposal, and that they have to guide customers less to the correct container.

And for the customers, it is nice that they do not have to lift their waste above the fence, they can receive a reward if they let their waste disposal be registered and they receive feedback and tutorials about the impact of the correct separation of their waste. This all hoping to improve their motivation, and in the end their behaviour.

Also, during the culture report made in Q1, group 12A concluded that there are multiple frustrations during the current procedure of the multiple Avalex locations (2). As stated in the introduction, these advantages tackle the following frustration themes: Information availability, attitude towards waste, impact and effort.

Feasibility

The Flipper also needs to be feasible for Avalex to implement the design on the different locations. The Avalex passes are already connected to the current database of Avalex, where they registered a pass (ID) to a home address. This database can easily be adapted to the NFC system of the Flipper, because the required personal information is already known to Avalex (7). The only thing for the users to do is accept that their amounts of weight are being registered. Avalex should keep in mind that the data needs to be secured very well, because customers' adresses are linked to the account.

The scaling platform is also not a problem. There are already comparable products that have a lift built around a scale (see figures 8-12).

The NFC reader will also not be a problem, there is an Arduino script made for the door of a recycle bin (6). The only thing that should be added is the digital scale that weighs the amount of waste and shows it on the intercom and registers it to the user's account. But that will not be too hard to add with an Arduino.

There should also be a separated button, for people that would not like to register the amount of waste to their account. Therefore, they will not receive a reward or any feedback.

There are already a few products on the market that also have a lift that throws away waste of lifts a container whre the waste is thrown in.



Figure 8-12: Products comparable to the Flipper.

Viability

The design is viable for Avalex because of better separation behaviour of customers, they will have less 'contaminated' containers. Which is helpful for the sorting centre. Maybe even saving money because less after separation is required.
For the users, it is also a viable product. When adapted to the Diftar system, it will save the customers of Avalex money to come and separate their waste.

The Flipper is also viable because when people come more often to Avalex because of their gain in knowledge or because of the Diftar-reward, they still have to pay for some materials. Therefore, Avalex can get more profit from certain materials when people visit Avalex more often.

The product itself is easy to maintain and clean. almost every part is made of aluminum or stainless steel, which can easily be cleaned with a water hose. Also, all elektricity cables can be installed beneath the supporting platform, which is easy to reach.

Next to that, when people separate more correctly, less help is needed from coworkers on the location. Which will allow Avalex to save money on salary.

But in the end, I do not know how people would actually respond to the Flipper and its interactive feedback/reward-system. People could come less to Avalex because they will separate their waste more at home due to the tutorials, or they come more often because of the Diftar-reward which lowers the waste taxes.

The manufacturing and welding costs are still missing in the cost analysis. But if we compare the Flipper to an underground container (including a Diftar system), the price of the underground container is 8000-9000 euro (9).
Unfortunately, the companies of the other comparable products did not want to share the prices of their products.



Figure 13: Underground Diftar containers.

For in the future, the Flipper can act as a sustainable substitute for incineration. The prices have gone up to incinerate, so processors are looking for more separated waste instead of residual waste (3).
Avalex can get a higher profit once their waste is more separated, with minimal contamination.

Cost Analysis

One of the criteria was that the design should be directly applicable, costing less then the given budget of 50.000 euros.

Certain costs like the website and the database are left out of the analysation because Avalex already have them operating. We only want to add some features to the existing service.

Part	Price (euro)
Flipper (stainless steel)	644
Hinge	30
Rotation motor	400
Shutter	305
Casing (aluminum)	1544
Weighing platform	129
Supporting platform	90
NFC reader	20
Screen	1100
Arduino	30
Container	320
Sensors	193
Total	4805

Figure 14: Cost calculation.

For the exact calculation, see Appendix D.
The overall price of the Flipper will be 3280, plus manufacturing costs, shipping costs and the costs covering the installation on the Avalex locations. When installed, the operational costs will be the electricity needed for the motor and interface, and the subscription of the database and website.

Iteration, Evaluation and Integration

Cultural

To make a product that provides feedback to the customers, I first needed to know which data is required to provide the right feedback for the desired behaviour.

For the research, we included people of all ages and lengths, because the Flipper should be accessible for every citizen in the municipality.

Feedback

As mentioned before, I wanted to achieve the desired customer behaviour through motivation. Therefore I have interviewed multiple people to identify what kind of feedback they would be interested in. The conclusion was that mostly, customers do not know a lot about their waste, the impact they have and to what kind of rocess they contribute (3).

The most common answers were knowing how much emissions they would have saved, what happens after the waste is collected at Avalex and if it would save them any money (appendix B).

These types of feedback are implemented into the website, except the emissions, which are also shown at the interface of the Flipper.

Reward

After interviewing an amount of volunteers, the answer for a motivator to come more often to Avalex was some kind of discount. Because Avalex cannot give away money to their customers (because they are subsidised by the municipalities) I had to come up with another reward.



At first, I wanted to use coupons, but not everyone is interested in a coupon for, as an example, a thrift store. While everyone benefits from a certain type of discount.

That is why I chose to adapt the Flippers data to the Diftar system, where the 'afvalstofheffing' or waste-taxes, are determined based on the amount of residual waste a household produces (8). When registering your data at the Avalex locations, you can get a discount on your taxes.
The Diftar system also works with a pass registered to a household.



Figure 19: The Diftar system.

I do believe that the given feedback will lead to a certain amount of cultural impact. Now it is comparable how much you have added to our circular environment. With all the influencers today, it could become a new trend to come to Avalex and show how much emissions you have saved in total.



Ergonomical

After the cultural goals had been set, I started conducting ergonomical research from where I could begin developing the structure and mechanics of the Flipper's product service system.

DINED

From the DINED database, two physical heights have been taken to research. Since there are two ways how people bring heavy waste to the container, an ellipse of both elbow height and hand height in a standing position was considered.

These heights have been taken under investigation during the ergonomical test. After identifying the correct height, it has been implented into the design.

To find out what is the best height for all people from Dutch society (20-94), who come to Avalex locations. The lowest value was chosen for this, to make the platform of the scale accessible to everyone in this way.

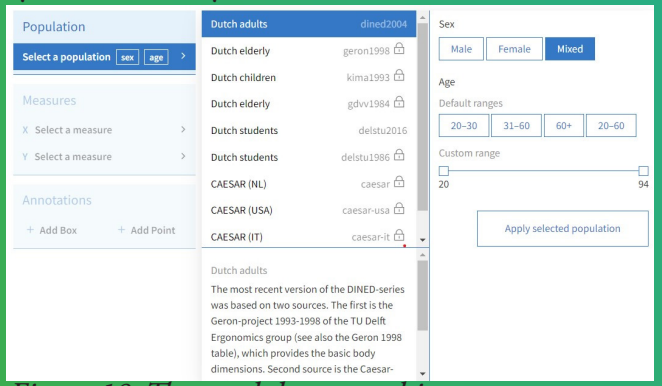


Figure 19: The used demographics.

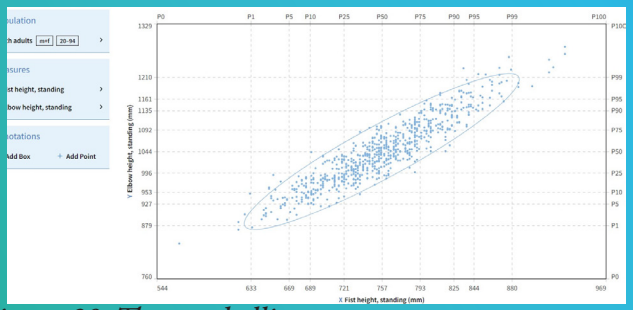


Figure 20: The used ellips.

The lowest values is 63.3 cm. This means that, to make the platform accessible to everyone, this becomes the height for the scale platform before the waste is thrown into the container.

Physical

The flipper is physically more comfortable for customers due to its lower platform, with a height of 63 cm. This is a lower height than the height of the fences at the Avalex locations. Not a lot, but for heavy products or for small/elderly people this is the preferred height compared to the elbow height of 88 cm. This has been concluded through the ergnomic evaluation test, see Appendix A.
Its outcome aligns with the fact that lifting requires more effort than putting waste down (10).
The only extra action customers have to perform is reaching the NFC reader/interface to shut the door.

Sensorical

Because the Flipper requires closed containers, the smell will also be reduced. The amount of noice will also be less because the sound of falling waste is muffled. And people are not able to see or reach the waste with a closed container. Which might lead to a more pleasant look at the different locations.

Cognitive

The interface has been made with as few buttons and options as possible, including an informative text explaining consequences of the two options of using a Avalex pass or the 'done'-button. The text is written in english so everyone can understand it. The interface has not been tested yet, so to minimize the chance of a misunderstanging, a letter should be send to all households, informing them about the usage of the Flipper.

Organisational

There will be less interactions between the employees of Avalex and the customers because they can find a lot of information online, and they do not need as much help with lifting waste as before. Also, the data is collected more precise and organised which is helpful for Avalex's daily business.

Behavioural

By combining all positive influences from the use of the Flipper, I hope that in the end, the behavioural ergonomics of the customers are improved.

Technological

After the ergonomic core values had been identified, the technological iteration of the Flipper could begin.1

Mechanism

One of the Flipper's features is that it throws away the waste into the container for the customer. But how can this be done best?
I looked at comparable products that also have a platform which lifts weight into another direction. From here I had two options, use a lifting table, or the hollow, triangled shaped surface.

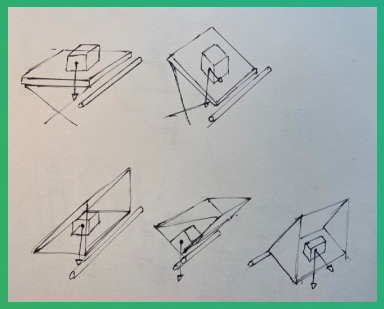


Figure 15: Lifting table vs. Flipper

During the technological test I concluded that the triangle shaped surface is more efficient, due to the fact that it does not experience as much friction resistance as the lifitng table, the centre of gravity of the waste is lifted across the fence, which results in a lower angle from where the weight starts falling down the container. The waste can also not fall down to the sides, which is safer for customers. From here, the Flipper got it's shape and mechanism.

The chosen hydraulic motor has 320 Nm. The radius will be 0,4m, because the centra of mass will be in the middle of the Flipper due too gravity.
Fmax= 320/0,2 = 1600N=163 kg which can be lifted and thrown away.

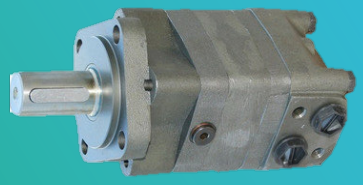


Figure 16: The chosen motor.

Interface

For the height of the interface, a study has been used where people were asked to scan their fingerprints at different heights (11). This height was comparable to the lowest elbow height of the DINED database. Also, Dutch people are a bit larger then people in America. Therefore, this height of 88 cm has been chosen.
For the screen, a wheater proof touchscreen monitor has been chosen to make the Flipper usable in every type of wheater.
For the software I have consulted an article (6).

Registration

To provide feedback and rewards after visiting Avalex, the amount of weight customer's dispose at Avalex needs to be registered to their account.
To achieve this, I chose a NFC system, which can be applied through either a NFC chip in the current Avalex passes, or through an extention on mobile phones (7).
The reason for a NFC system is because it does not require any actions other than holding the chip near the interface. Which is faster compared to other systems as Bluetooth for instance.
It is also not dependend on a direct line of sight compared to Infrared.
It's system consumes very little power, which is ideal for long daily usage at the locations.

Supporting platform

To support all weight from the Flipper, including its scaling platform and the weight of the waste, a strong, stainless steel table has been chosen.
The Flipper weighs 7950* ((0,004*0,6*0,45)+ (0,004*0,6*0,4) + 0,004*0,45*0,4)) = 22 kg. The selected table can handle a total weight of 400 kg, to prevent the Flipper from breaking.



Figure 17: The supporting platform made of stainles steel.

Door

To make sure that the right amount of waste is registered on te customer's account, a door is applied to make sure that there is no possibility to throw away extra waste or take away waste from the flipper once it has registered the amount on an account.

First, the idea was that the flipper was closed at the customer's arrival. But this would require an extra step to first scan the pass or press the button to open the door, then walk back to your car to pick up waste and then press the button again to close the door.
This was also done to prevent people from disposing waste and then leave without registration.

But after consulting an expert on IT and data protection law, this changed. Because customers must be informed with a consent form about the usage of their data. If they do not agree, there must also be an option to not register your waste disposal (1). Therefore, I chose to have an open door.

Also, I chose a shutter door to prevent the door from blocking any large waste products. It works fast, is automated and does not add a large amount

Weighing system

The amount of waste needs to be weighted before linking it to the customer's account.
Therefore I have chosen to implement a scaling surface on the bottom platform of the Flipper.
This platform is made out of two stainless steel sheets, with nine compression sensors between them for a good representation of the amount of weight on the entire surface.

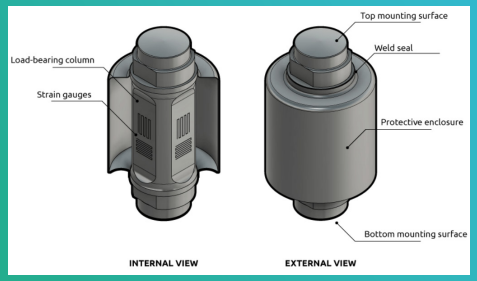




Figure 18: The weighing platform.

To measure the amount of weight, the mean of the total amount of sensors is calculated. After the Flipper is rotated back to its original position, the scale calibrates.

Casing

For the casing of the flipper it is necesary to make sure that it is strong, easy to clean and aesthetically more attrative than the current concrete context of the Avalex locations.

Therefore, I have chosen a green coloured casing, for the 'green' behaviour performed at Avalex, which is made out of made out of aluminum. This can be welded together to fit the area around the Flipper and its supporting platform.

Conclusion & Client Advice

Firstly, I do believe that this kind of feedback/rewarding system about the impact of the customers' visit at Avalex can help them achieve their own vision. The services granted through the website are helpful, educative, insightful and save money.

All stakeholders benefit from the implementation of the Flipper.

Because they are installed on the walking route of the locations, it limits the amount of space where people can park their car. This is a disadvantage of the Flipper design.

The design is feasible. It can be made, but the costs are probably higher than the estimated cost calculation. This is because the amount and price of required labour necessary for its installation is not precise.
Also, the costs of the adaptations of the Avalex locations, (like breaking up roads, installing electricity etc.) have not been taken into account. Before the Flipper can be applied, this should be determined.

If we take 48 hours x 20 euro = 960 euro for manufacturing costs, and then the installment of again around 1000 euros. Prices to install electricity around the location differ, this can reach 4500 euros in big projects like this one. Now we are already around 11.500 euro for one Flipper. Which is not too bad compared to the underground containers for 9000 euro.

The estimated amount of time to install a Flipper without construction adaptations of the Avalex location is one week. To place, install and test all features including electricity and road changes takes a minimum of 2-3 weeks, depending on the amount of employees that are going to be hired. With manufacturing and shipping time the estimated time will be a month.

To realise the usage of the Flipper, we must bring the amount of Flippers that are going to be installed back to 4. Only making it necessary for the materials that Avalex is interested in.
These materials that need to be thrown away by using a Flipper can change per period. For instance, each month Avalex posts new materials that you can bring to the locations to weight them and receive a reward or feedback.

To start off the societal impact Avalex needs to send a letter to all citizens to inform them about the new PSS. This can also take more than a week before all letters have been send and delivered to the adresses.
Also, the Avalex passes need to be upgraded with a NFC chip. Which will be around 100.000 euro, 1 euro per NFC pass per household, for all 100.000 households in Rijswijk, Delft and Wassenaar. Therefore, an NFC extention on mobile phones is a more cheaper option, but this will not be as easy to install for everyone in our society.
So, it is a big investment, but once up and running the system can provide digital communication of valuable data for all stakeholders.

Also, not all types of feedback and information need to be dependent on the Flipper. If Avalex wants to change customers' behaviour they can already update their app and website with tutorials and a materials guide where to throw away which material.

But to hand out rewards and feedback about the impact customers have, you will need the data gathered through the Flipper.

My final advice is to first start off with a pilot to test how customers will actually respond to the feedback and rewards. And to see if customers start to separate their waste more correctly at the locations. After a positive pilot, analyse how much the profits have been raised during the pilot period, and calculate the amount of operational months Avalex has to make before they can afford a Flipper. If that number is above 6 months, Avalex should compare the profits against the costs to decide if a Flipper is viable enough for them to apply.

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Appendix A: Concept evaluation test

H1 Research aim

The 'Flipper' is meant to measure the amount of waste people throw away from a certain material at a time. This must be a relevant amount of waste and for a relevant material which can be reused again. Therefore I am planning to implement the Flipper only on heavy materials(banden, bouw- en sloopafval, dakleer, gips, (geïmpregneerd) hout, grof huishoudelijk afval, metalen en puin).

Research question:
Is the chosen hand heights for the weighing platform ergonomically preferred over the elbow height? (63 or 88 cm).

Hypothesis:
I expect the outcome to be that the lowest heights are the best heights. This is because when people are carrying an amount of waste which they can easily handle, it will be no problem to lay the waste down to 63 cm. And if they are carrying heavy objects it would be nice to not have to lift the products over the fences to throw them away in the container, but to have a platform a bit lower which is less physically demanding.

The ergonomic subdomain this research focuses on considering physical ergonomics is the perception of effort and comfort when carrying and laying down weight (from car to Flipper/container).

H2 Method

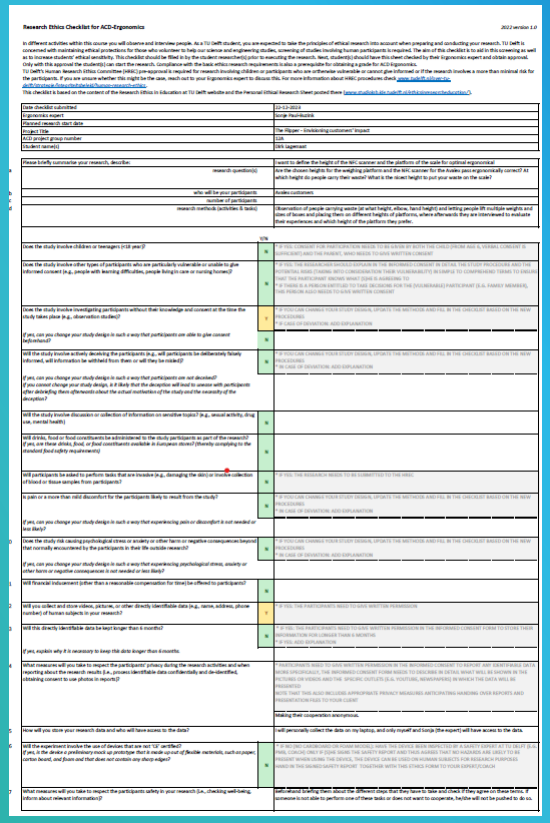
This research will consist of two different research methods. One research will be focused on the height people carry their waste. The other one will be more focused on the most acceptable height of the platform where the waste needs to be put on.

Demographics:
For the socio-demographic part of my research, I need to include all customers that can come to Avalex, which is everyone in our society. The recycling street is a public space which is available for everyone.

Recruitment:
Ask people in my own environment if they want to cooperate in the study. Or go to one of the Avalex locations and observe customers while they bring their waste from their cars towards the containers.

I am specifically looking for people from different heights. This is because the height of the weighing platform should be tested, and it should be representative for both short and long people. This is to make the final design accessible for everyone.

Research ethics & informed consent form:



Research Ethics Checklist for ACD-Ergonomics 2022 version 1.0

In different activities within this course you will observe and interview people. As a TU Delft student, you are expected to take the principles of ethical research into account when preparing and conducting your research. TU Delft is concerned with maintaining ethical protections for those who volunteer to help our science and engineering studies, screening of studies involving human participants is required. The aim of this checklist is to aid in this screening as well as to increase students' ethical sensitivity. This checklist should be filled in by the student researcher(s) prior to executing the research. Next, student(s) should have this sheet checked by their ergonomics expert and initial approval. Only with this approval the student(s) can start the research. Compliance with the basic ethics research requirements is also a prerequisite for obtaining a grade for ACD-Ergonomics.
TU Delft's Human Research Ethics Committee (HREC) pre-approval is required for research involving children or participants who are otherwise vulnerable or cannot give informed consent. If the research involves a more than minimal risk for the participants, if you are unsure whether this might be the case, reach out to your ergonomics expert to discuss this. For more information about HREC procedures check www.tudelft.nl/over-tu-delft/strategie/integriteit/human-research-ethics.
This checklist is based on the content of the Research Ethics in Education at TU Delft website and the Personal Ethical Research Sheet posted there (www.studiegids.tudelft.nl/default.aspx?page=course&id=ae4c7).

Date checklist submitted	22-12-2022
Ergonomics expert	Sonja Pauchlová
Planned research start date	
Project title	The Flipper – Envisioning customers' impact
ACD project group number	12A
Student name(s)	Dirk Jagersvel

Please briefly summarize your research, describe:	
research question(s)	I want to define the height of the WC scanner and the platform of the scale for optimal ergonomical. Are the chosen heights for the weighing platform and the WC scanner for the Avalex pass ergonomically correct? At which height do people carry their waste? What is the nicest height to put your waste on the scale?
who will be your participants	Avalex customers
number of participants	
research methods (activities & tasks)	Observation of people carrying waste (at what height, elbow, hand height) and letting people lift multiple weights and sizes of boxes and placing them on different heights of platform, where afterwards they are interviewed to evaluate their experiences and which height of the platform they prefer.

		YES
Does the study involve children or teenagers (<18 year)?	N	IF YES: CONSENT FOR PARTICIPATION NEEDS TO BE GIVEN BY BOTH THE CHILD (FROM AGE 6, VERBAL CONSENT IS SUFFICIENT) AND THE PARENT, WHO NEEDS TO GIVE WRITTEN CONSENT
Does the study involve other types of participants who are particularly vulnerable or unable to give informed consent (e.g., people with learning difficulties, people living in care or nursing homes)?	N	IF YES, THE RESEARCHER SHOULD EXPLAIN IN THE INFORMED CONSENT IN DETAIL THE STUDY PROCEDURE AND THE POTENTIAL RISKS (TAKING INTO CONSIDERATION THEIR VULNERABILITY) IN SIMPLE TO COMPREHEND TERMS TO ENSURE THAT THE PARTICIPANT KNOWS WHAT (S)HE IS AGREEING TO IF THERE IS A PERSON ENTITLED TO TAKE DECISIONS FOR THE (VULNERABLE) PARTICIPANT (E.G. FAMILY MEMBERS), THIS PERSON ALSO NEEDS TO GIVE WRITTEN CONSENT
Does the study involve investigating participants without their knowledge and consent at the time the study takes place (e.g., observation studies)?	N	IF YOU CAN CHANGE YOUR STUDY DESIGN, UPDATE THE METHODS AND FILL IN THE CHECKLIST BASED ON THE NEW PROCEDURE IF CASE OF DEVIATION: ADD EXPLANATION
If yes, can you change your study design in such a way that participants are able to give consent beforehand?	N	
Will the study involve actively deceiving the participants (e.g., will participants be deliberately falsely informed, will information be withheld from them or will they be misled)?	N	IF YOU CAN CHANGE YOUR STUDY DESIGN, UPDATE THE METHODS AND FILL IN THE CHECKLIST BASED ON THE NEW PROCEDURE IF CASE OF DEVIATION: ADD EXPLANATION
If yes, can you change your study design in such a way that participants are not deceived? If you cannot change your study design, is it likely that the deception will lead to unease with participants after debriefing them afterwards about the actual motivation of the study and the necessity of the deception?		
Will the study involve discussion or collection of information on sensitive topics? (e.g., sexual activity, drug use, mental health)	N	
Will drinks, food or food constituents be administered to the study participants as part of the research? *If yes, are these drinks, food, or food constituents available in European stores? (thereby complying to the standard food safety requirements)*	N	
Will participants be asked to perform tasks that are invasive (e.g., damaging the skin) or involve collection of blood or tissue samples from participants?	N	IF YES, THE RESEARCH NEEDS TO BE SUBMITTED TO THE HREC
Is pain or a more than mild discomfort for the participants likely to result from the study?	N	IF YOU CAN CHANGE YOUR STUDY DESIGN, UPDATE THE METHODS AND FILL IN THE CHECKLIST BASED ON THE NEW PROCEDURE IF CASE OF DEVIATION: ADD EXPLANATION
If yes, can you change your study design in such a way that experiencing pain or discomfort is not needed or less likely?		
Does the study risk causing psychological stress or anxiety or other harm or negative consequences beyond that normally encountered by the participants in their life outside research?	N	IF YOU CAN CHANGE YOUR STUDY DESIGN, UPDATE THE METHODS AND FILL IN THE CHECKLIST BASED ON THE NEW PROCEDURE IF CASE OF DEVIATION: ADD EXPLANATION
If yes, can you change your study design in such a way that experiencing psychological stress, anxiety or other harm or negative consequences is not needed or less likely?		
Will financial inducement (other than a reasonable compensation for time) be offered to participants?	N	
Will you collect and store videos, pictures, or other directly identifiable data (e.g., name, address, phone number) of human subjects in your research?	Y	IF YES: THE PARTICIPANTS NEED TO GIVE WRITTEN PERMISSION
Will this directly identifiable data be kept longer than 6 months?	N	IF YES: THE PARTICIPANTS NEED TO GIVE WRITTEN PERMISSION IN THE INFORMED CONSENT FORM TO STORE THEIR INFORMATION FOR LONGER THAN 6 MONTHS IF YES: ADD EXPLANATION
If yes, explain why it is necessary to keep this data longer than 6 months		
What measures will you take to respect the participants' privacy during the research activities and when reporting about the research results (i.e., process identifiable data confidentially and de-identified, obtaining consent to use photos in reports)?		IF PARTICIPANTS NEED TO GIVE WRITTEN PERMISSION IN THE INFORMED CONSENT TO REPORT ANY IDENTIFIABLE DATA MORE SPECIFICALLY, THE INFORMED CONSENT FORM NEEDS TO DESCRIBE IN DETAIL WHAT WILL BE SHOWN IN THE PICTURES OR VIDEOS AND THE SPECIFIC OUTLETS (E.G. YOUTUBE, NEWSPAPERS) IN WHICH THE DATA WILL BE PRESENTED NOTE THAT THIS ALSO INCLUDES APPROPRIATE PRIVACY MEASURES ANTICIPATING HANDING OVER REPORTS AND PRESENTATION FILES TO YOUR CLIENT Making their cooperation anonymous.
How will you store your research data and who will have access to the data?		I will personally collect the data on my laptop, and only myself and Sonja (the expert) will have access to the data.
Will the experiment involve the use of devices that are not 'CE' certified? *If yes, is the device a preliminary mock up prototype that is made up out of flexible materials, such as paper, carton board, and foam and that does not contain any sharp edges?*	N	IF NO (AND ARDUINOS OR FOAM MODELS): HAVE THE DEVICE BEEN INSPECTED BY A SAFETY EXPERT AT TU DELFT (SEE PME, COACHE ONLY F [DOE DAAN THE SAFETY REPORT AND THEY AGREED THAT NO HAZARDS ARE LIKELY TO BE PRESENT WHEN USING THE DEVICE, THIS DEVICE CAN BE USED ON HUMAN SUBJECTS FOR RESEARCH PURPOSES HAND IN THE SIGNED SAFETY REPORT TOGETHER WITH THIS ETHICS FORM TO YOUR EXPERT/COACH
What measures will you take to respect the participants safety in your research (i.e., checking well-being, inform about relevant information)?		Both attend briefing them about the different tasks that they have to take and check if they agree on these tasks. If someone is not able to perform one of these tasks or does not want to cooperate, he/she will not be pushed to do so.

Wat zijn fysiek ergonomisch de beste hoogten voor de werking van de 'Flipper'?

Dit onderzoek wordt uitgevoerd als onderdeel van de MSc opleiding Industrieel Ontwerpen aan de TU Delft.
Studenten: Dirk Langemaat
Contactpersoon: [Dirk Langemaat , D.C.R.Lagemaat@student.tudelft.nl +31 6 28605684]

Toestemmingsverklaring participant

Ik neem vrijwillig deel aan dit onderzoek.
Ik erken dat ik vooraf voldoende informatie en uitleg heb gekregen over dit onderzoek en al mijn vragen zijn naar voldoening beantwoord. Ik heb de tijd gekregen die ik nodig had om in te stemmen met de deelname. Op elk moment kan ik vragen stellen met betrekking tot het onderzoek.

Mij is bekend dat dit onderzoek bestaat uit:

1. Observatie
2. Diepgaand interview (20 min)
3. Foto's/video's/stemopnames maken voor gebruik in het onderzoek.

Ik ben mij ervan bewust dat tijdens het onderzoek gegevens worden verzameld in de vorm van bijvoorbeeld aantekeningen, foto's, video's en/of geluidsopnames. Ik geef toestemming voor het verzamelen van deze gegevens en het maken van geluidsopnames, foto's en video opnames tijdens het onderzoek. Gegevens zullen geanonimiseerd worden verwerkt en geanalyseerd (zonder naam of andere identificeerbare informatie). Deze gegevens zijn alleen voor het onderzoeksteam en hun TU Delft begeleiders beschikbaar.

De foto's, video's en/of geluidsopnames zullen worden gebruikt ter ondersteuning van het analyseren van verzamelde gegevens. Video opnames en foto's kunnen tevens worden gebruikt ter illustratie van onderzoeksbevindingen in publicaties en presentaties over het project.

Ik geef toestemming voor het gebruik van foto's en video opnames van mijn deelname:
(selecteer wat van toepassing is)

☐ waarin ik herkenbaar ben voor publicaties en presentaties over het project.

☐ waarin ik niet herkenbaar ben voor publicaties en presentaties over het project.

☐ enkel voor data analyse doeleinden en niet voor publicaties en presentaties over het project.

Ik geef toestemming om gegevens nog maximaal 5 jaar na afloop van dit onderzoek te bewaren en te gebruiken voor onderwijs- en onderzoeksdoeleinden.

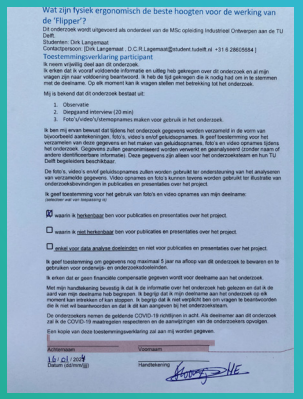
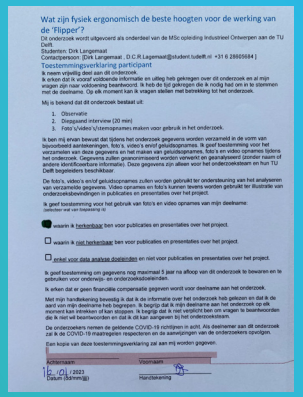
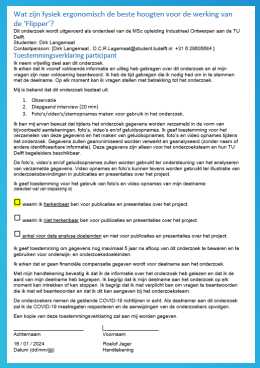
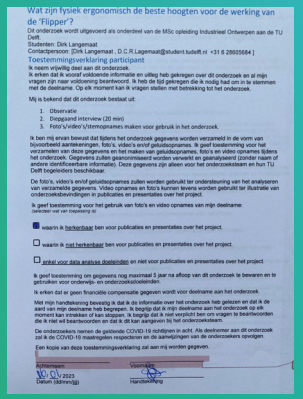
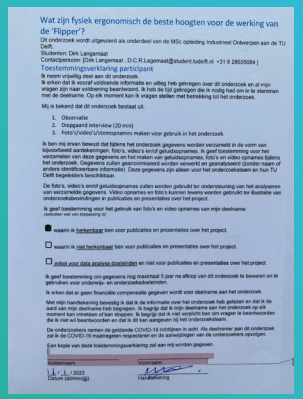
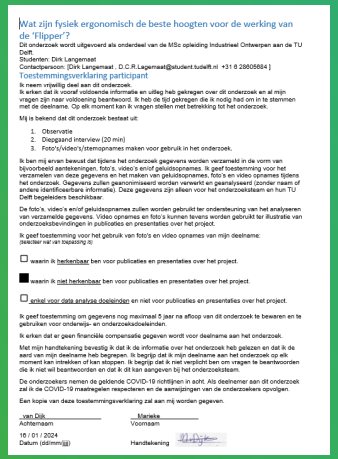
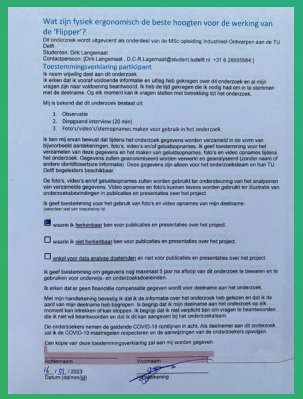
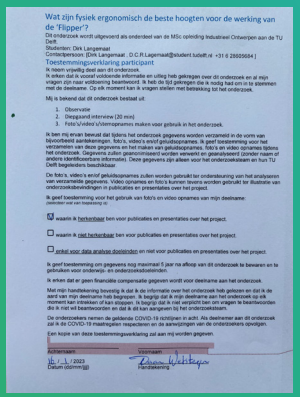
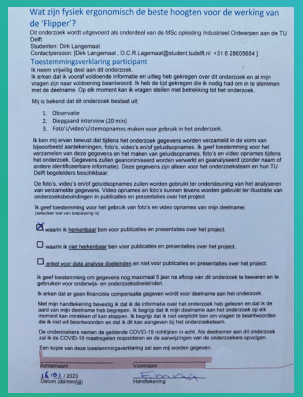
Ik erken dat er geen financiële compensatie gegeven wordt voor deelname aan het onderzoek.

Met mijn handtekening bevestig ik dat ik de informatie over het onderzoek heb gelezen en dat ik de aard van mijn deelname heb begrepen. Ik begrijp dat ik mijn deelname aan het onderzoek op elk moment kan intrekken of kan stoppen. Ik begrijp dat ik niet verplicht ben om vragen te beantwoorden die ik niet wil beantwoorden en dat ik dit kan aangeven bij het onderzoeksteam.

De onderzoekers nemen de geldende COVID-19 richtlijnen in acht. Als deelnemer aan dit onderzoek zal ik de COVID-19 maatregelen respecteren en de aanwijzingen van de onderzoekers opvolgen.

Een kopie van deze toestemmingsverklaring zal aan mij worden gegeven.

Achternaam	Voornaam
16 / 01 / 2024	Handtekening
Datum (dd/mm/jjjj)	

Equipment & Tools

For this experiment I need to have:

- A platform which can simulate the platform of the scale.
- Weights to simulate how customers carry different amounts of weights.
- Different sizes of boxes to put the weights in and simulate smaller/bigger objects. (A carton box of 32 cm long and 48 cm in with, and a plastic box of 34 x 52 cm).
- A (video)camera for examples of people carrying and placing boxes.





Procedure

Observation during test:
Looking at carrying heights with different weight loads.
Check how people lay the waste down on different platform heights. Specifically, watch the posture and people's backs.

The physical research can be done at any place, it does not need to be executed at one of the Avalex locations.

The volunteers are asked to execute the same procedure multiple times with different loads and weights.
They are asked to move the box, filled with weights, from a height of 70 cm (mean height car boot) towards a platform (differs from 63 and 88 cm). Afterwards they are asked to answer the questionnaire regarding the activities.

Interviewing:
After cooperation of the simulated environment a questionnaire with questions about what people find the most acceptable height for all different kinds of weights and sizes of simulated waste.

Interview questionaire

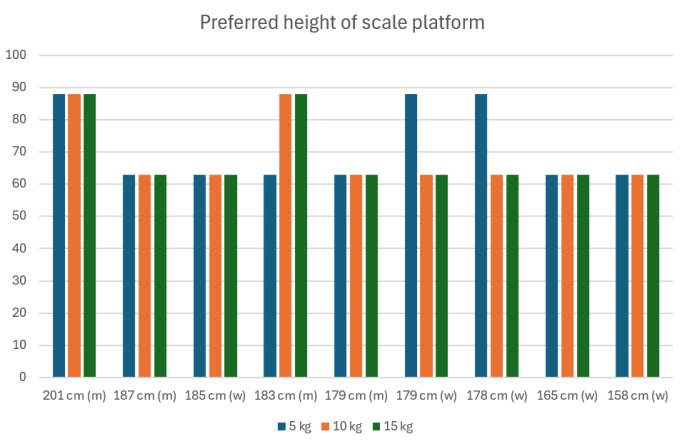
- Do you have a background in designing, researching or psychology? y/n
- What is your length?
- What posture did you find most comfortable to adopt while lifting and carrying the different sizes and loads of weight? (elbow/hand height)
- What height did you find most comfortable to put the 'waste' on?
- Do you ever have trouble lifting heavy products?
- Would you be interested in knowing how much CO2 emissions you have saved by throwing away certain types of materials at the Avalex locations?
- What would motivate you (what kind of reward/feedback) to come more often to Avalex?

Data processing & Analysis

From the results, different preferences are measured and put into a table and graph to visualise the preferred height.

Results

From the results, different preferences are measured and put into a table and graph to visualise the preferred height.



Person	Gender	Length	Age	Size of box	Amount of weight	Preferred posture	Preferred height of platform
1	man	201 cm	22	32 x 48 cm	5 kg	elbow height	88
					10 kg	elbow height	88
					15 kg	hand height	88
2	Woman	185 cm	18	32 x 48 cm	5 kg	elbow height	63
					10 kg	hand height	63
					15 kg	hand height	63
3	Man	183 cm	20	32 x 48 cm	5 kg	elbow height	63
					10 kg	elbow height	88
					15 kg	elbow height	88
4	Man	179 cm	19	32 x 48 cm	5 kg	elbow height	63
					10 kg	hand height	63
					15 kg	hand height	63
5	woman	179 cm	18	32 x 48 cm	5 kg	elbow height	88
					10 kg	elbow height	63
					15 kg	elbow height	63
6	Woman	165 cm	21	34 x 52 cm	5 kg	elbow height	63
					10 kg	elbow height	63
					15 kg	hand height	63
7	Woman	158 cm	23	34 x 52 cm	5 kg	elbow height	63
					10 kg	elbow height	63
					15 kg	elbow height	63
8	Man	187 cm	23	34 x 52 cm	5 kg	hand height	63
					10 kg	hand height	63
					15 kg	hand height	63
9	Woman	178 cm	19	34 x 52 cm	5 kg	elbow height	88
					10 kg	hand height	63
					15 kg	hand height	63

Discussion

During the concept evaluation test, there was a brief moment where the different volunteers could see each other. Even though this moment was not during the different actions that were performed during the test, this could have affected the outcome. In the future this can be resolved by having the next volunteer wait in a separate room.

Conclusion

As shown in the results, in 74,07% of the actions the volunteers preferred the platform height of 63 cm. When carrying heavy weight (15 kg) this percentage rises to 77,78%.
Also, the only persons who preferred the height of 88 cm were male, and 183 and 201 cm long. Which is not representative for our whole society.

And in these cases, they would have to put less extra effort in bending forwards then when shorter people have to lift the weight higher. It is easier to lower waste from elbow height to 63 cm than to lift it to 88 cm.

So to make the Flipper accessible for everyone in our community and to assure a minimum of required effort, the height of 63 cm has been chosen as the height for the scale platform of the Flipper.

Appendix B: Technological test

Research aim

To make a choice between the lifting table and the 'Flipper', I completed a test between the two designs. The test is focused on the ease of throwing away the waste after the scale has measured and registered the amount of waste after the customer has scanned their pass.

With the lifting table, the scaling platform would be pushed up at the backend by a hydraulic pump. This would create a slope from which the waste is led down. This will create friction between the two surfaces.
At the Flipper on the other hand, the waste should first turn 90 degrees, from this point the waste its centre of gravity is pushed over the fence which would need a smaller angle.

So the research question is: 'Which design requires a smaller angle to throw away different types of materials?'

Method

For this experiment I made a prototype.
It consists of wood, multiplex, with a thin layer of aluminium to represent the stainless steel platform of the Flipper.
With the use of different weights (5, 10 & 15kg) I tested from which angle different types of materials (wood, metal, carton) started to fall down into the 'container'. The angles have been measured by using a drafting triangle.

Procedure

During the test, I will first lift the platform of the lifting table, untill all the 'waste' has fallen off. After, I will do the same with the Flipper prototype to compare the angles when the 'waste' has been thrown away.



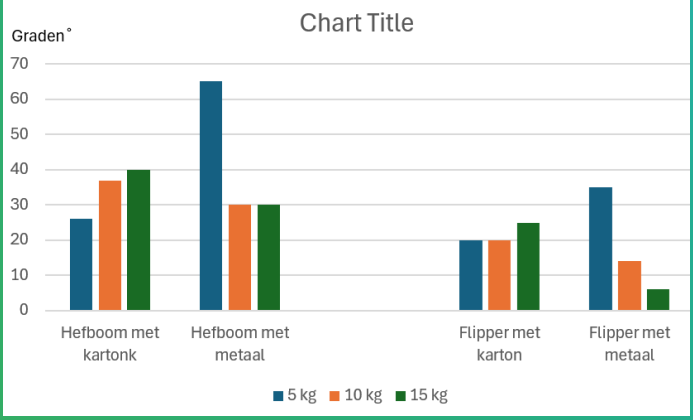






Results

As shown in the graph, after the Flipper turend 90 degrees, it required a smaller angle to let the waste fall down.



Discussion & Conslusion

For my prototype I used an aluminium surface. This on the other hand is not the material I am planning to adapt in the final design. Stainless steel was not used due to financial incapabilities.

How smooth certain types of materials with different amounts of waste fall down, also depends on their shape and surface. There can not be a standard angle below 60 degrees with the lift table (heftafel).

With the lifting table, not all waste fell down at the same time.
With the Flipper, you would require a smaller angle, therefore it is more efficient to use it when dealing with the same amount and type of waste as the lifting table.

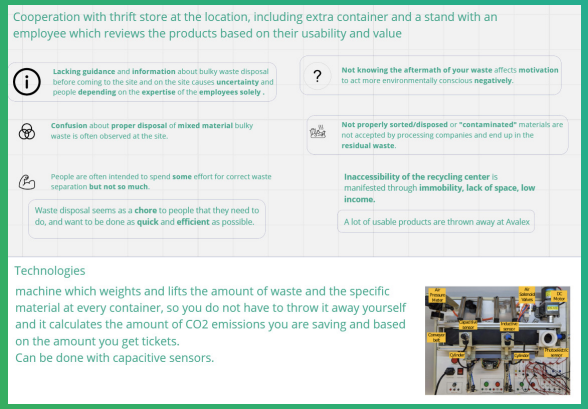
During the interview after the experiment, the volunteers were asked if they would be motivated to visit Avalex again after receiving feedback about the impact of their visit. Almost every volunteer said yes, but to test if the overall PSS works, we need to test it in real life, with an actual working interface and website, where the amount of emissions saved by the users is shown, and check when they return for a follow up visit to Avalex.

On the other hand, when following the guidance of separation-at-home-videos, this could also lead to a smaller number of people visiting Avalex because of the tips provided through the website.

Appendix C: Design Process

Link to Miro board: https://miro.com/app/board/uXjVNR3Tf08=/?share_link_id=667255566580

I started with 4 ideas, cooperating with thrift stores, upgrading the app for a carpool service, a help screen for mixed materials and the elevator/conveyor belt.



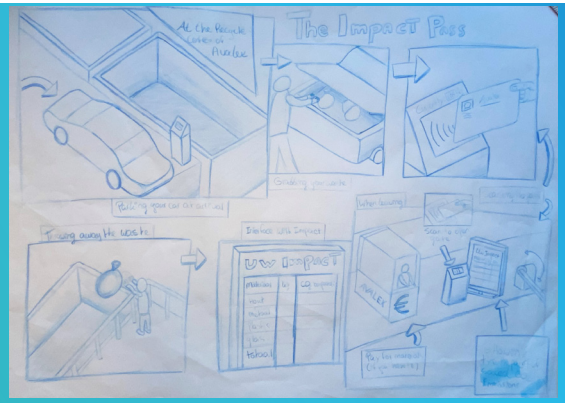




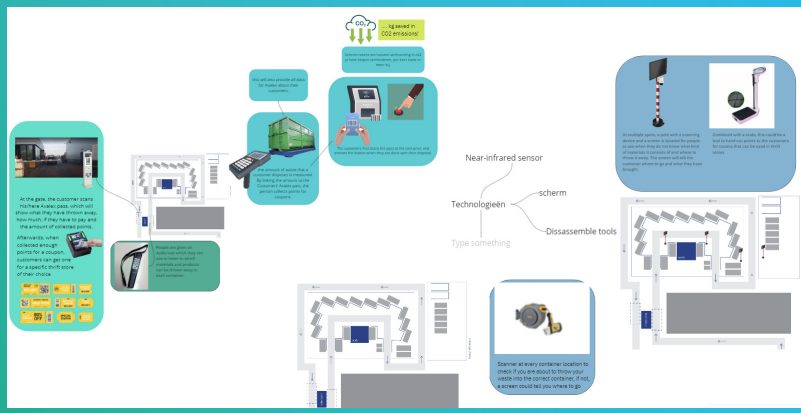


Almost from the beginning of the second period of the ACD project, I was interested in some sort of rewarding system.

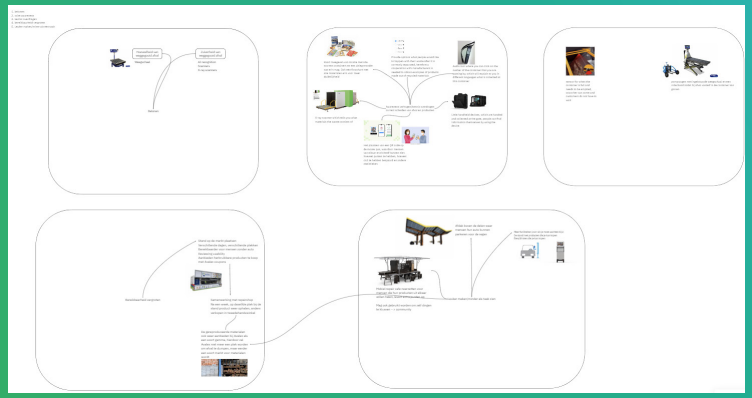




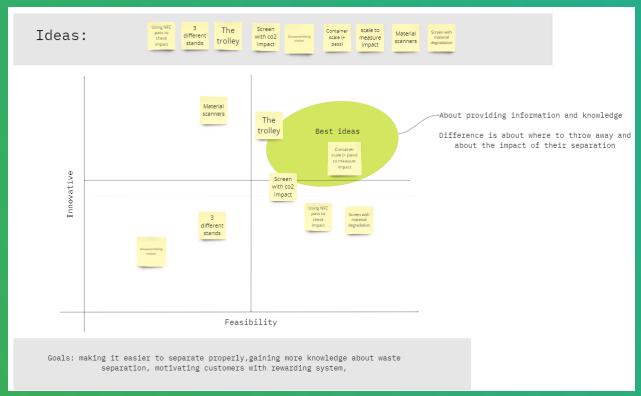
From here I started to do some How-To's or HKJ's in Dutch. This one is about sharing knowledge.



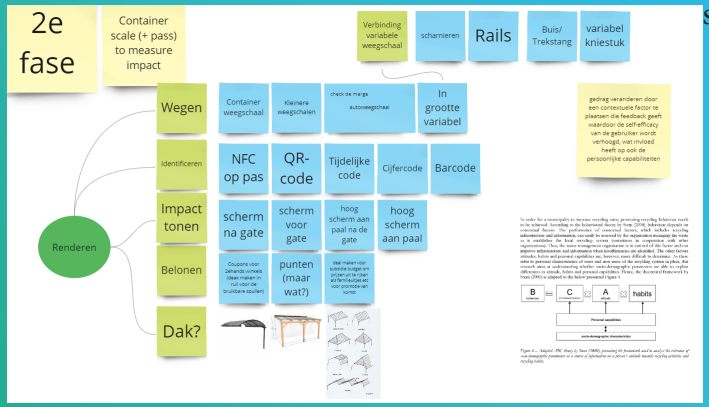
And these are How-To's about rewarding, raising awareness, sharing knowledge, increase accessibility and making the visits more fun/less as task.

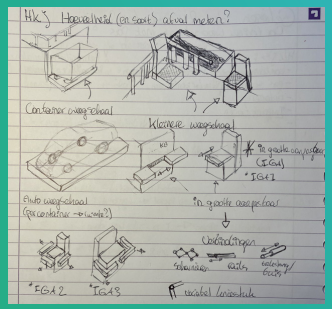


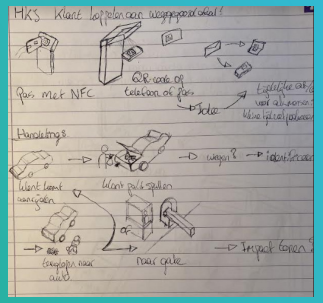
From this point, i started writing down ideas and put them in a C-box to select the most relevant ideas based on their innovative and feasibility aspects.

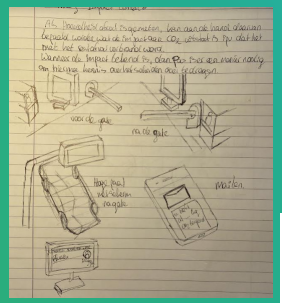


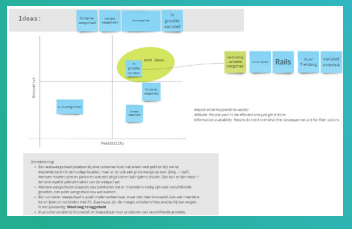
Next, I chose the most valuable idea and split it up in different sub categories. For these categories I also made some How-To's.

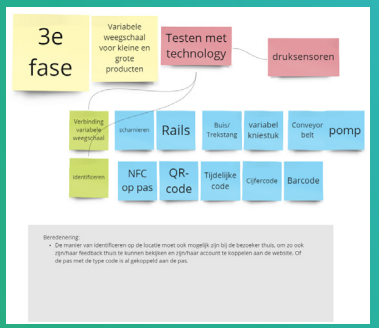




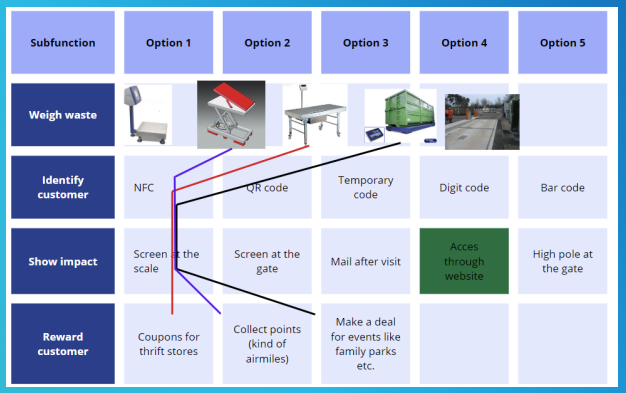




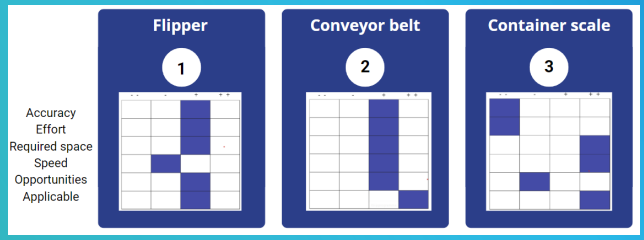


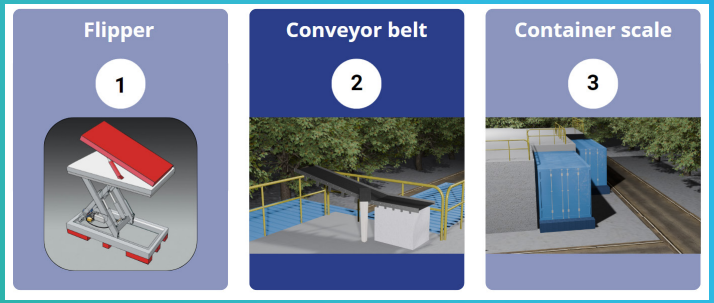


After all options were identified, I made a morphological chart to come up with 3 different concepts: The Flipper, the Conveyor belt and the Container scale.



I used the Harris profile method to decide which concept was best. And the previous outcome was that the conveyor belt was the best option.

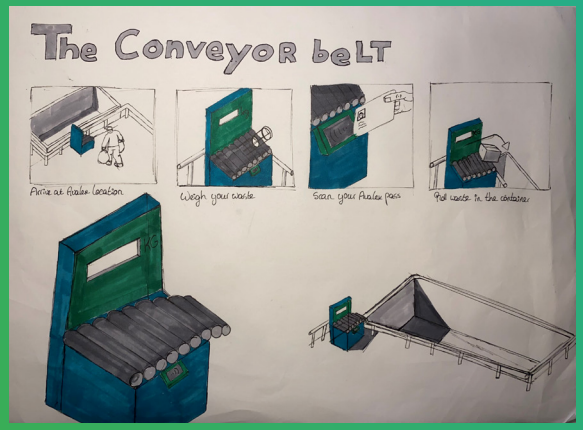


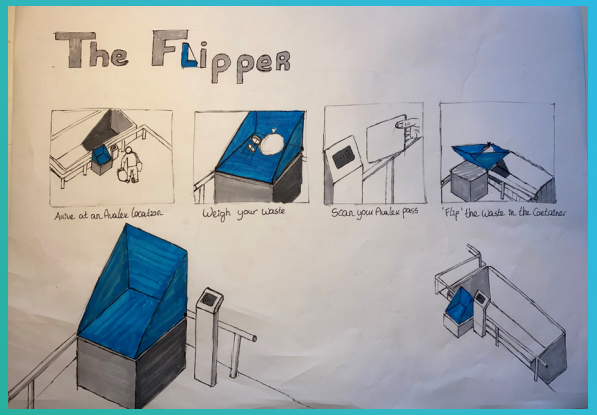


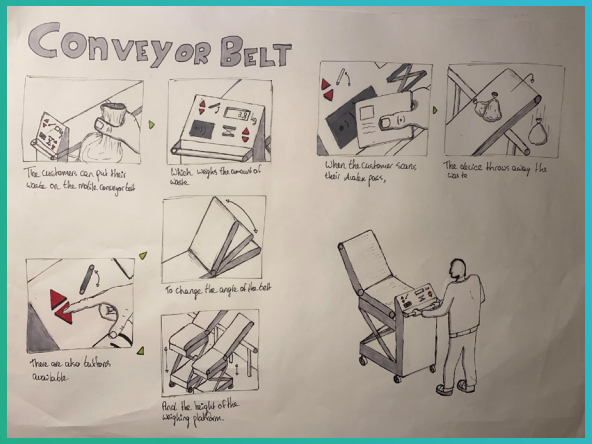
After talking to the client and my coach, I have decided to not continue with the conveyor belt, but to iterate more on the Flipper.

The conveyor belt has some limits. It requires more space at the Avalex location than the Flipper.
The waste can roll downhill, back to the customer, which is not safe. Especially with heavy products, where a lot of grip is needed.
The conveyor belt is made with a rubber platform, which is hard to clean.
And it would be harder to implement a weighing scale.

So then, I started to look at all the different actions that are required by a customer to use the Flipper, I was uncertain about is the Flipper should be mobile or fixed on a location.



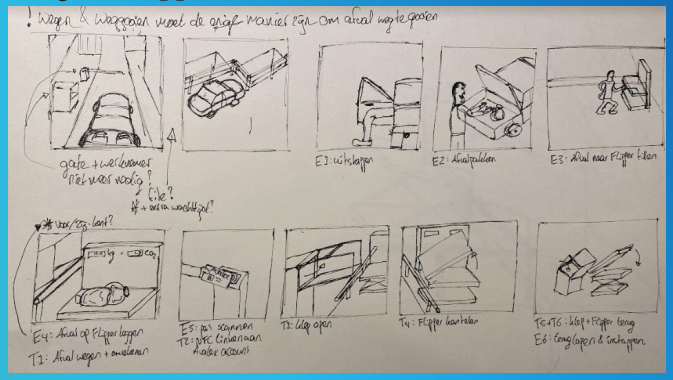




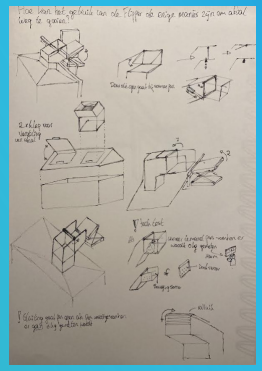
But in the end, I chose a fixed location for the Flipper because it requires less time and physical effort, there are no customers walking around the limited space area with mobile trolley's, less physical actions are required to throw away the waste, and a closed container is easier, else there should be an opening which can recognise if a customer puts the conveyor belt near the container.

The only positive point is that the customers do not have to cary the waste from their car to the container.

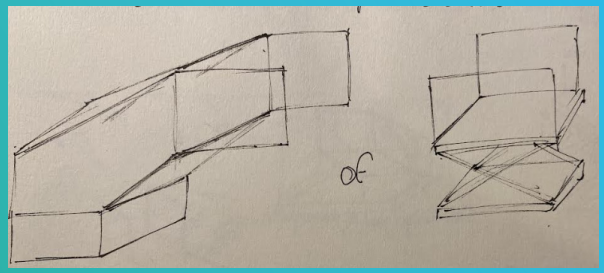
After this decision, I made a user journey, to optimise the different steps that have to be taken when using the Flipper.

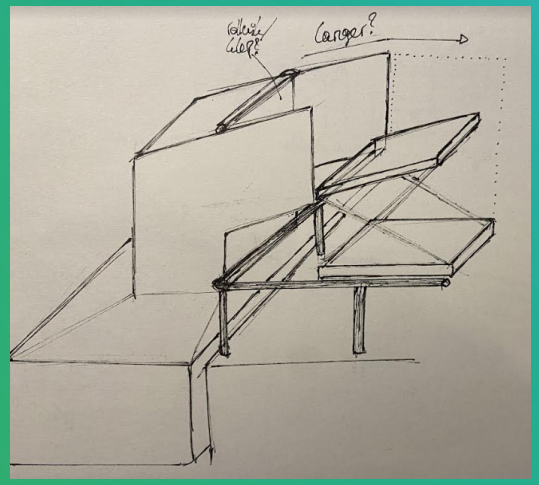


To prevent customers to throw away waste without using the Flipper I made another How-To:

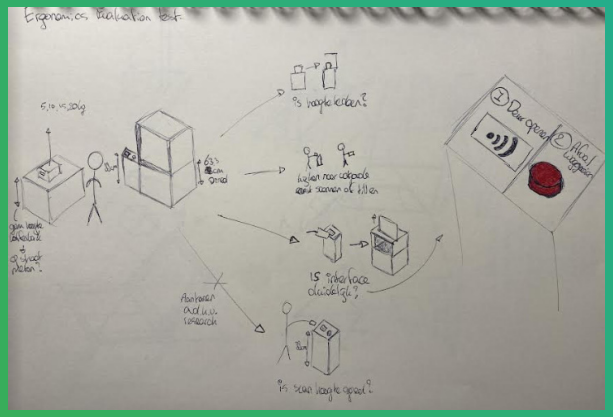


And to provide maximum safety, I chose an embodiement which covers the total design.



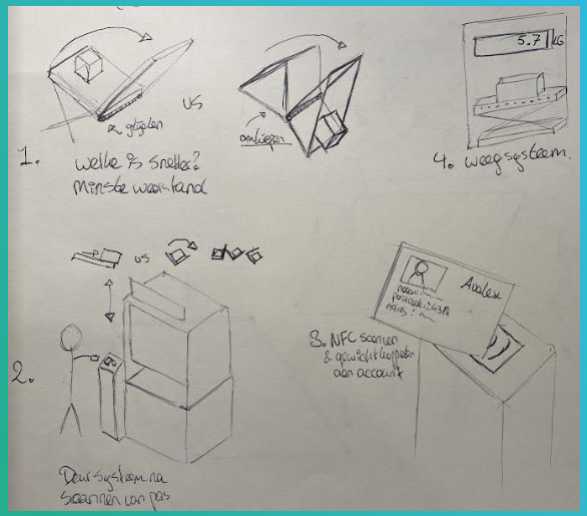


First I had to decide what ergonomic test I wanted to do. So I made several options:

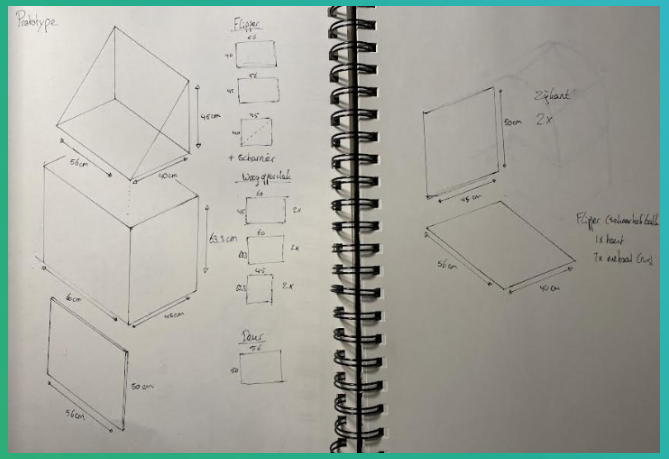


I chose by comparing the options and thinking to myself: 'What is crucial to know for the funtionality of the Flipper?', which is the height of the platform where the waste is put on by the customer.

After the height was determined, I did the same with the technolocial uncertainties, from which the mechanism to throw away the waste was the most important one.

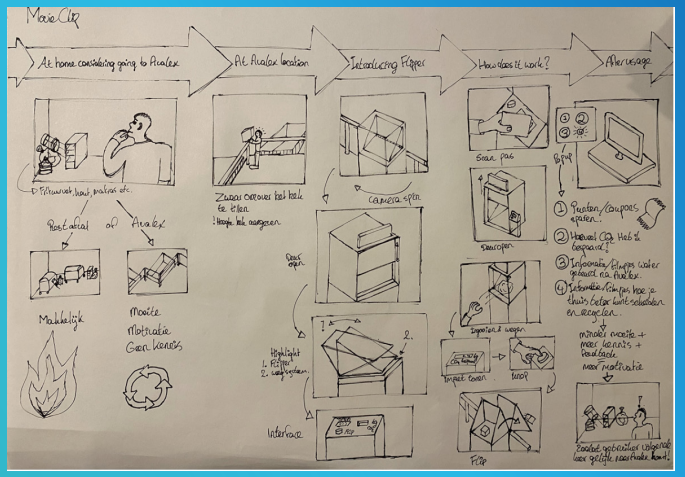


These are the drawings and dimensions I used to make a prototype for the ergonomical and technological test.



After determining the height and type of mechanism I had all required information for the further development of the Flipper.

This is my idea for the movieclip.



Appendix D: Cost Analysis

Part	Price (euro)
Flipper (stainless steel)	644
Hinge	30
Rotation motor	400
Shutter	305
Casing (aluminum)	1544
Weighing platform	129
Supporting platform	90
NFC reader	20
Screen	1100
Arduino	30
Container	320
Sensors	193
Total	4805

To calculate the total costs of the Flipper, I used prices of comparable parts, and calculated the materials by multiplying the quantity of materials by the selling price.

The total price for the parts is 4805 euros.

Flipper: 1 stainless steel plate of 2000x1000x4 mm is 644 euro's. https://roestvast.nl/product/rvs-plaat-316/
Hinges: aluminum made with nuts for 10 mm bolts is 4 x 7,50 = 30 euro's. https://www.systeal.com/en/other-accessories/560-anodized-aluminium-hinge-for-10-mm-slot-40x40-profile.html
Rotation motor: https://www.hydrauliek24.nl/m-s-100cc-hydraulische-motor-32-mm-as?utm_source=-google-surfaces&utm_medium=organic&gad_source=1&gclid=CjwKCAiAzc2tBhA6EiwArv-i6WdQP-zCND2kQcwHfn9QsS_1UFi_WDMj_-uiN3J7j_epu76ZwOJ4VPBoCkEYQAvD_BwE
Shutter: Including the correct simensions and extra options for cables. https://www.rolluiken33.nl/alu-tech-basisline-rolluik.html?gclid=Cj0KCQiAwbitBhDIARIsABfFYIJaEHmPW_nBccOv7RsbSAJsyn-pXOS2F1_4u5fH3agePzGgfuH-d-a8aAvE1EALw_wcB
Casing: around 3,8 m2 of aluminum is necessary, but to be safe I estimated 4 m2. https://onlinealuminium.nl/gewalste-plaat-en-aw-6082-t651/?utm_source=Google%20Shopping&utm_campaign=Feed%20Online%20Aluminium&utm_medium=cpc&utm_term=3030&gad_source=1&gclid=CjwKCAiAzc2tBhA6EiwArv-i6SorA9bGP9PwXS8bCM5UO80g6EDzbSRyVwRY1mPlOyc9fko6CeXk9RoCEWYQAvD_BwE
Weiging platform, 2x (0,002x0,6x0,45) * kostprijs: https://metaalstore.nl/rvs-304-plaat-geslepen/?gclid=Cj0KCQiAqsitBhDlARIsAGMR1Rjk4AIGnZ91mFQfXXBvw64eThv5g0GFakAtPsAEwGpquUpIldgSEoMaAoY2EALw_wcB
Supporting platform: https://www.ggmgastro.com/nl-nl-eur/roestvrijstalen-werktafel-eco-600x600mm-met-onderplank-zonder-opstaande-rand-atk66-eco?gad_source=1&gclid=CjwKCAiAzc2tBhA6EiwArv-i6S0QjldnRTXmvFND4p1AF7e04wMfRzddijdGtbtzN5NhLlSBLvKVIhoCL3YQAvD_BwE
NFC reader: https://ar.vittascience.com/shop/43/rfid-reader-125-khz-grove
Screen: https://www.touchwindow.com/c/MimoMOD-10180CH.html
Arduino: https://www.amazon.nl/Arduino-A000066-Uno-REV3/dp/B008GRTSV6/ref=asc_df_B008GRTSV6/?tag=nlshogostdde-21&linkCode=df0&hvadid=430533494283&hvpos=&hvnetw=g&hvrand=16574251724597687334&hvpone=&hvptwo=&hvqmt=&hvdev=c&hvdvcmdl=&hvlocint=&hvlocphy=1010704&hvtargid=pla-457497319401&psc=1&mcid=ce088103de893fdc8a92c350d0ff6f8f
Container: https://www.afzetbak.nl/container-huren/bouw-en-sloopafval/10m3-gesloten?utm_source=-google&utm_medium=feed&utm_campaign=product-bouw-en-sloopafval-10m3gesloten&gad_source=1&gclid=CjwKCAiA5L2tBhBTEiwAdSxJX1y5rEi7aSFYGMMXv2S1d1YrQ3RHd7MEOoAfl-hhJLMzb8PDVzq0EnxoC1OkQAvD_BwE#augmented-reality
Sensors: 9 * 21,42. https://nl.rs-online.com/web/p/pressure-sensor-ics/8212431?cm_mmc=NL-PLA-DS3A-_-google-_-PLA_NL_NL_Semiconductors_Whoop+(2)-_-(NL:Whoop!)+Pressure+Sensor+ICs-_-8212431&matchtype=&pla-343619750335&gad_source=1&gclid=CjwKCAiAzc2tBhA6EiwArv-i6cQcorym0rWVn98kLJm-RnPZysx9Bk6ruEiMy-TEQb019o9viqWvFRoCdlIQAvD_BwE&gclsrc=aw.ds